

The Attack: How it works

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Content

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The Attack: How it works

Every time you open LinkedIn in a Chrome-based browser, LinkedIn's JavaScript executes a silent scan of your installed browser extensions. The scan probes for thousands of specific extensions by ID, collects the results, encrypts them, and transmits them to LinkedIn's servers. The entire process happens in the background. There is no consent dialog, no notification, no mention of it in LinkedIn's privacy policy.

This page documents exactly how the system works, with line references and code excerpts from LinkedIn's production JavaScript bundle.

Source File

All code references on this page come from a single JavaScript bundle served to every LinkedIn visitor. The filename is a content hash that changes with each deployment (e.g. `5fdhwcppjcvqvxawd8pg1n51.js`), but the stable identifiers are:

```
| Property | Value | | Webpack chunk ID | chunk.905 | | Extension scan module | 75023 | |  
| Bundle size | ~2.7 MB | | Framework | Ember.js  
(globalThis.webpackChunk_ember_auto_import_) | |
```

The bundle is a Webpack package containing multiple modules. Three of those modules form the scanning system described below.

Line numbers referenced on this page are from the December 2025 version of the bundle. They may shift between deployments, but the code structures, string literals, and module exports remain searchable by keyword.

Architecture

LinkedIn's extension detection consists of three cooperating systems inside the same JavaScript bundle:

| System | Internal Name | Function | | APFC / DNA | `triggerApfc`, `triggerDnaApfcEvent` | Device fingerprinting engine. Collects 48 browser characteristics. | | AED | `AedEvent`, `fetchExtensions` | Active Extension Detection. Probes for known extensions using `fetch()`. | | Spectroscopy | `SpectroscopyEvent`, `scanDOMForPrefix` | Passive extension detection. Scans the DOM for evidence of extension activity. | |

All three systems feed into the same telemetry pipeline: LinkedIn's `li/track` endpoint.

The Extension List

At line 9571, character offset 443, inside Webpack module `75023`, there is a hardcoded array:

```
const r = [
  {id: "aaaeoelkococjpgngfokhbkkfiiegolp", file: "assets/index-C0XueBxP.js"},
  {id: "aabfjmnamliahmlicgeogldnfaaklfon", file: "images/logo.svg"},
  {id: "aacbpaggdjcbngmgjgpkpddliddineni", file: "sidebar.html"},
  // ... thousands more entries ...
];
```

Each entry has two fields:

- `id`: A 32-character Chrome Web Store extension ID
- `file`: A known file path inside that extension's package, such as `popup.html`, `icon.png`, or `manifest.json`

The `file` field is not incidental. Someone at LinkedIn has identified a specific internal resource for each extension that is declared as web-accessible. This is the probe target.

As of December 2025, the array contained **5,459 entries**. By February 2026, it had grown to **6,167**. The array alone occupies roughly 409,000 characters of source code.

Stage 1: Active Extension Detection (AED)

AED is a brute-force scan. It attempts to load a known file from each extension using the `fetch()` API.

Chrome extensions can expose internal files to web pages through the `web_accessible_resources` field in their `manifest.json`. When an extension is installed and has exposed a resource, a `fetch()` request to `chrome-extension://{id}/{file}` will

succeed. When the extension is not installed, Chrome blocks the request and the promise rejects.

LinkedIn tests every extension in the list this way.

Method 1: Parallel batch scan

Lines 9573 to 9576:

```
async function c() {
  const e = [],
    t = r.map(({id: t, file: n}) => {
      return fetch(`chrome-extension://${t}/${n}`)
    });
  (await Promise.allSettled(t)).forEach((t, n) => {
    if ("fulfilled" === t.status && void 0 !== t.value) {
      const t = r[n];
      t && e.push(t.id);
    }
  });
  return e;
}
```

This fires all 6,222 `fetch()` requests simultaneously using `Promise.allSettled()`. Every request that resolves as `"fulfilled"` means that extension is installed. The function returns an array of detected extension IDs.

Method 2: Staggered sequential scan

Lines 9578 to 9579:

```
async function(e) {
  const t = [];
  for (const {id: n, file: i} of r) {
    try {
      await fetch(`chrome-extension://${n}/${i}`) && t.push(n);
    } catch(e) {}
    e > 0 && await new Promise(t => setTimeout(t, e));
  }
  return t;
}
```

This alternative probes extensions one at a time with a configurable delay between each request. Failed fetches are silently caught and discarded. The delay parameter

(`staggerDetectionMs`) allows LinkedIn to throttle the scan, reducing its visibility in network monitoring tools and lowering CPU impact.

Which method runs

Lines 9577 to 9579:

```
const {
  useRequestIdleCallback: i = false,
  timeout: o = 2000,
  staggerDetectionMs: l = 0
} = n;

const d = async () => {
  const n = l > 0
    ? await staggeredScan(l) // Method 2
    : await c();             // Method 1
  // ... fire tracking events ...
};

i && "function" == typeof window.requestIdleCallback
  ? window.requestIdleCallback(d, {timeout: o})
  : await d();
```

If `staggerDetectionMs` is greater than zero, LinkedIn uses the slower sequential scan. Otherwise it fires the parallel batch. The scan can also be deferred to `requestIdleCallback`, which delays execution until the browser is idle. The user sees no performance impact. The scan leaves no visible trace.

Stage 2: Passive DOM Scanning (Spectroscopy)

Independent of the AED scan, LinkedIn runs a second detection system that walks the entire DOM tree looking for evidence of extension activity.

Many Chrome extensions inject elements into web pages: modified HTML, added scripts, altered attributes, UI overlays. When they do, the injected content often contains references to the extension's internal URL scheme (`chrome-extension://`). Spectroscopy finds these references.

Implementation

Lines 9581 to 9587:

```

const d = "chrome-extension://";

function p(e, t, n) {
  // Scan text nodes
  if (e.nodeType === Node.TEXT_NODE
    && void 0 !== e.textContent
    && e.textContent.includes(t))
    n.push(u(e.textContent));

  // Scan element attributes
  if (e.nodeType === Node.ELEMENT_NODE)
    for (let i = 0; i < e.attributes.length; i++) {
      const o = e.attributes.item(i);
      void 0 !== o.value && o.value.includes(t) && n.push(u(o.value));
    }

  // Recurse into child nodes
  for (let i = 0; i < e.childNodes.length; i++)
    p(e.childNodes[i], t, n);
}

```

The function `p` starts at the document root and recursively inspects every node. For text nodes, it checks whether the text contains `chrome-extension://`. For element nodes, it checks every attribute value. When it finds a match, it extracts the 32-character extension ID from the URL.

The results are fired as a `SpectroscopyEvent`:

```

Array.isArray(n) && n.length > 0
  && e.fireTrackingPayload("SpectroscopyEvent", {browserExtensionIds: n, ...t});

```

Why two detection methods

| Method | Technique | What it catches | | | AED | `fetch()` against known resource paths |
 Extensions that are merely installed, even if they inject nothing into the current page | | |
 Spectroscopy | Full DOM tree walk | Extensions that actively modify the page, even if they
 are not in LinkedIn's hardcoded list | |

AED requires a pre-built target list but can detect passive extensions. Spectroscopy requires no list but only catches extensions that leave traces in the DOM. Together, they cover both cases.

Data Transmission

Detected extension IDs are sent to LinkedIn's servers through two tracking events.

AedEvent (active scan results)

Line 9580:

```
Array.isArray(n) && n.length > 0  
&& e.fireTrackingPayload("AedEvent", {browserExtensionIds: n, ...t});
```

SpectroscopyEvent (passive scan results)

Line 9587:

```
Array.isArray(n) && n.length > 0  
&& e.fireTrackingPayload("SpectroscopyEvent", {browserExtensionIds: n, ...t});
```

Both events carry a `browserExtensionIds` array containing the Chrome extension IDs detected on that user's browser. Both feed into the same telemetry transport.

The telemetry pipeline

The `fireTrackingPayload` method sends data to LinkedIn's `li/track` endpoint:

```
https://www.linkedin.com/li/track
```

This endpoint is referenced at lines 2026, 4288, and 11785. The transport layer batches up to 29 events per request (line 2098), retries up to 4 times on failure, and supports LZ-based compression before transmission (function `compressToBase64`, lines 9587 to 9601).

Encryption

Before transmission, the fingerprint payload (which includes the extension scan results) is encrypted.

Line 9528:

```
f.encryptWithKeyFromDifferentSources(
  JSON.stringify(t),
  "apfcDfPK",    // Public key identifier
  "apfcDfPKV",  // Public key version
  n, r
).then(t => {
  globalThis.apfcDf = t;
  // ... transmit to telemetry endpoint ...
});
```

The payload is serialized to JSON, encrypted using an RSA public key identified as `apfcDfPK`, and stored on `globalThis.apfcDf`. From there, it is transmitted to two endpoints:

- `/platform-telemetry/li/apfcDf`
- `/apfc/collect`

The encrypted fingerprint is also injected as an HTTP header into subsequent API requests made during the user's session (via `SyncCollectionHandler`, line 9525). This means the fingerprint does not get sent once. It accompanies every API call the user makes for the duration of their visit.

Browser Targeting

Lines 9572 to 9577:

```
function a() {
  return "undefined" !== typeof window
    && window
    && "node" !== window.appEnvironment;
}

function s() {
  return window?.navigator?.userAgent?.indexOf("Chrome") > -1;
}

if (!a() || !s()) return;
```

The extension scan runs only in Chrome-based browsers. The `isUserAgentChrome()` function checks for “Chrome” in the user agent string. The `isBrowser()` function excludes server-side rendering environments. If either check fails, the scan does not execute.

This means every user visiting LinkedIn with Chrome, Edge, Brave, Opera, Arc, or any other Chromium-based browser is subject to the scan.

The Larger Fingerprinting System: APFC

The extension scan is one component of a broader device fingerprinting system called APFC (Anti-fraud Platform Features Collection), internally also referred to as DNA (Device Network Analysis).

APFC collects 48 distinct browser characteristics (line 2260):

#	Feature	What it collects
1	<code>webrtc</code>	Local IP address via WebRTC
2	<code>enumerateDevices</code>	Connected cameras, microphones, speakers
3-6	<code>appName</code> , <code>tsSeed</code> , <code>appVersion</code> , <code>appCodeName</code>	Browser identification strings
7	<code>location</code>	Page URL components (protocol, hostname, port, origin, href, hash, pathname)
8	<code>javascripts</code>	JavaScript engine characteristics
9-13	<code>platform</code> , <code>product</code> , <code>productSub</code> , <code>cpuClass</code> , <code>oscpu</code>	OS and CPU identification
14	<code>hardwareConcurrency</code>	Number of CPU cores
15	<code>deviceMemory</code>	Device RAM in GB
16-17	<code>vendor</code> , <code>vendorSub</code>	Browser vendor strings
18	<code>language</code>	Browser language
19-20	<code>timezoneOffset</code> , <code>timezone</code>	Timezone data
21	<code>userAgent</code>	Full user agent string
22	<code>webdriver</code>	Whether the browser is controlled by automation
23	<code>doNotTrack</code>	Do Not Track setting
24	<code>incognito</code>	Whether the user is in private browsing mode
25-30		Screen properties Color depth, pixel depth, pixel ratio, resolution, orientation, available resolution
31-35		Storage detection <code>sessionStorage</code> , <code>localStorage</code> , <code>IndexedDB</code> , <code>addBehavior</code> , <code>openDatabase</code>
36	<code>canvas</code>	Canvas fingerprint (rendered hidden element, hashed)
37	<code>webgl</code>	WebGL renderer, vendor, extensions, 65+ parameter values
38	<code>signals</code>	Browser lie detection (spoofed OS, browser, resolution, language)
39	<code>touchSupport</code>	Touch screen capabilities
40	<code>networkInfo</code>	Connection type, downlink speed, round-trip time
41	<code>battery</code>	Battery level, charging status, estimated discharge time
42	<code>audio</code>	AudioContext fingerprint (oscillator, compressor, analyser nodes)
43	<code>automation</code>	Automation framework detection
44-45	<code>plugins</code> , <code>mimetypes</code>	Browser plugin list and MIME types
46-47	<code>fonts</code> , <code>fontsFlash</code>	Installed system fonts
48	<code>getFeatures</code>	Meta-feature: the collection process itself

Feature #23 is worth noting in isolation. LinkedIn collects the user's Do Not Track preference, then excludes it from the fingerprint hash (line 9512, `excludes: { doNotTrack: true }`). They record that you asked not to be tracked. Then they track you.

Anti-Detection Design

Several implementation choices reveal that this system was designed to avoid detection:

Idle execution. The scan can be deferred to `requestIdleCallback`, which runs the code only when the browser has no other work to do. The user sees no performance degradation, no spinning indicator, no delay.

Staggered probing. The sequential scan mode introduces a configurable delay between each `fetch()` request, spreading thousands of network requests over time instead of firing them in a single burst that might appear in developer tools or network monitors.

Hidden iframe. The HUMAN Security (PerimeterX) integration loads a hidden iframe from `li.protechts.net` that is 0 by 0 pixels, positioned at `left: -9999px`, and marked `aria-hidden="true"` (lines 9536 to 9552).

Silent error handling. Failed `fetch()` requests are caught with empty `catch` blocks. No errors are logged to the console. No warnings are shown.

RSA encryption. The fingerprint payload is encrypted before transmission. Even if a user inspects the network request in developer tools, the payload contents are not readable.

Third-Party Integrations

The APFC system also feeds data to three external services:

HUMAN Security (formerly PerimeterX)

Lines 9536 to 9552:

```
c = "https://li.protechts.net/index.html?ts=" + r
  + "&r_id=" + encodeURIComponent(o)
  + "&app_id=" + a + "&uc=scraping"
```

LinkedIn loads a hidden iframe from `li.protechts.net`, passing a timestamp, the page's tree ID, a hashed session cookie (`bcookie`), and the app ID (`PXd0jV695v` in production). The iframe reads and sets PerimeterX cookies (`_px3`, `_pxhd`, `_pxvid`, `pxcts`) via cross-origin `postMessage`.

Merchant Pool (Device Fingerprinting)

Lines 9529 to 9535:

```
const n = `https://merchantpool1.linkedin.com/mdt.js
?session_id=${bcookie}&instanceId=${instanceId}`
```

A separate fingerprinting script is loaded from `merchantpool1.linkedin.com`, passing the user's session cookie and a hardcoded instance ID (`fb6bbd47-fa7c-4264-b4e9-b25948407586`).

Google reCAPTCHA v3 Enterprise

Lines 9553 to 9560. LinkedIn loads `https://www.google.com/recaptcha/enterprise.js` and executes it on page load with action `"onPageLoad"`, collecting the resulting token.

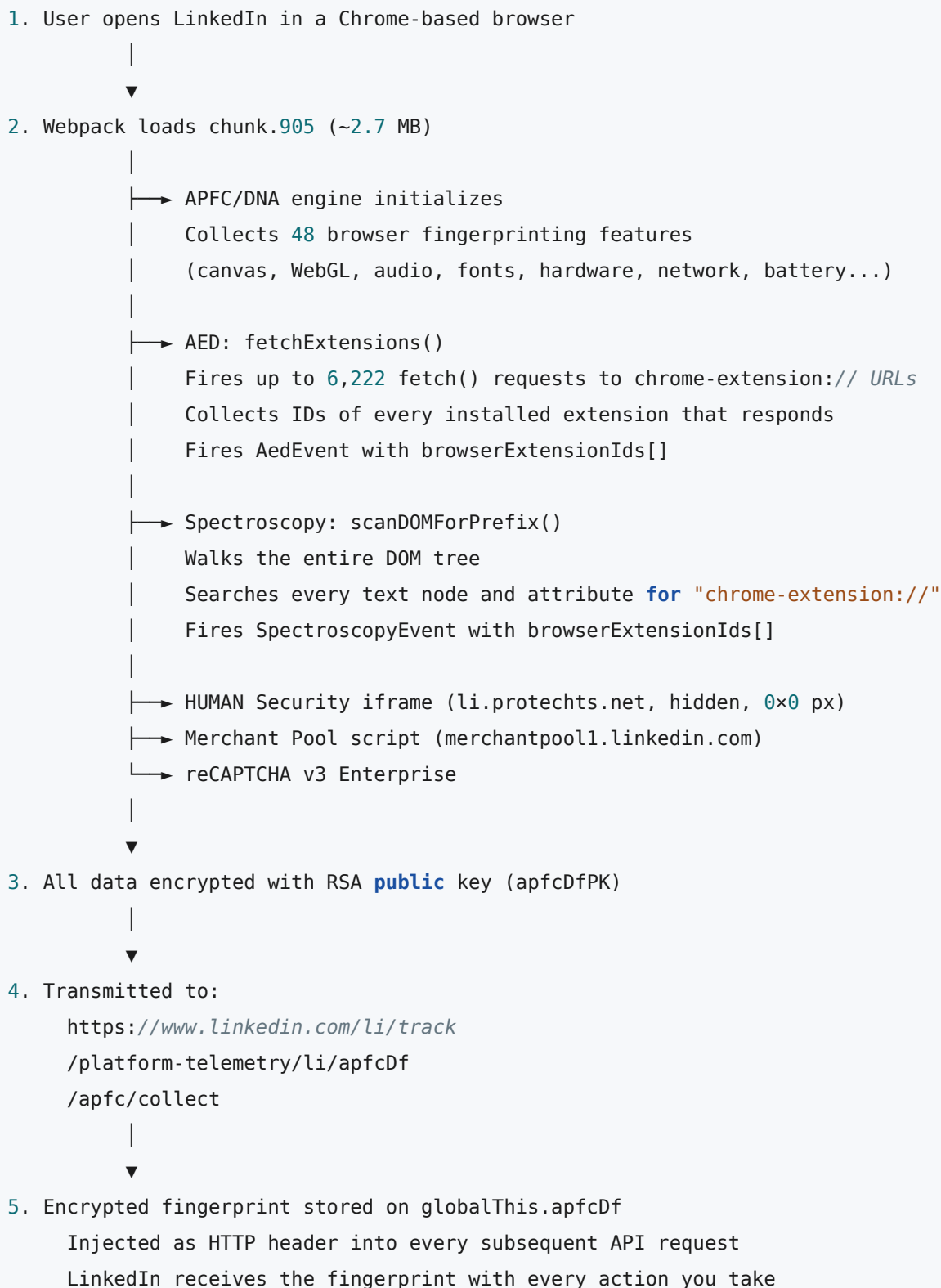
Feature Flags

The fingerprinting and scanning systems are controlled by LinkedIn's internal experimentation platform, LIX (LinkedIn Internal eXperimentation):

Flag	Controls		
<code>pemberly.tracking.fireApfcEvent</code>	DNA fingerprint collection		
<code>pemberly.tracking.human.integration</code>	HUMAN Security integration		
<code>pemberly.tracking.dfp.integration</code>	Merchant Pool fingerprinting		
<code>pemberly.tracking.recaptcha.v3</code>	reCAPTCHA v3 integration		
<code>pemberly.tracking.apfc.network.interceptor</code>	Network request interception		
<code>pemberly.web.ondemand</code>	On-demand fingerprinting mode		
<code>sync.apfc.headers</code>	Sync fingerprint via HTTP headers		
<code>sync.apfc.couchbase</code>	Sync fingerprint to Couchbase		
<code>fingerprinting.collection.skip.performance.marker.check</code>	Skip performance marker check		

These flags allow LinkedIn to enable or disable fingerprinting for specific user segments, run A/B tests on scanning behavior, and roll out changes incrementally. The existence of A/B testing flags for a surveillance system means LinkedIn is actively experimenting with how to scan users more effectively.

Data Flow Summary



How to Verify This Yourself

All of the above can be verified independently.

- Open LinkedIn in Chrome.
- Open Developer Tools (F12).

- Go to the Network tab.
- Filter by `.js` and look for the largest JavaScript bundle (~2.7 MB), or search the page source for `chunk.905`.
- Open the file. Search for `chrome-extension://`. You will find the extension array in module `75023`.
- Search for `AedEvent` and `SpectroscopyEvent` to find the tracking event triggers.
- Search for `apfcDfPK` to find the encryption logic.
- Search for `li/track` to find the telemetry endpoint.
- Watch the Network tab for POST requests to `https://www.linkedin.com/li/track` after the page loads.

The code is minified and partially obfuscated, but the string literals, endpoint URLs, module exports, and control flow are preserved. Everything documented on this page can be read directly from the source.

Growth Rate

| Date | Extensions in the scan list | | 2017 | 38 | | 2024 | ~461 | | May 2025 | ~1,000 | | December 2025 | 5,459 | | February 2026 | 6,167 | |

LinkedIn added 708 extensions to the scan list between December 2025 and February 2026. That is roughly 12 new extensions per day. The system is not static. It is actively maintained and expanding.

All line numbers and code excerpts reference LinkedIn's production JavaScript bundle (Webpack chunk.905, module 75023) as served to Chrome users. The filename is a content hash that changes with each deployment. The file is minified, but variable names in module exports, string literals, endpoint URLs, and control flow structures are sufficiently preserved to reconstruct the system described above.

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